

FIRE: Mid-Air Thermo-Tactile Display

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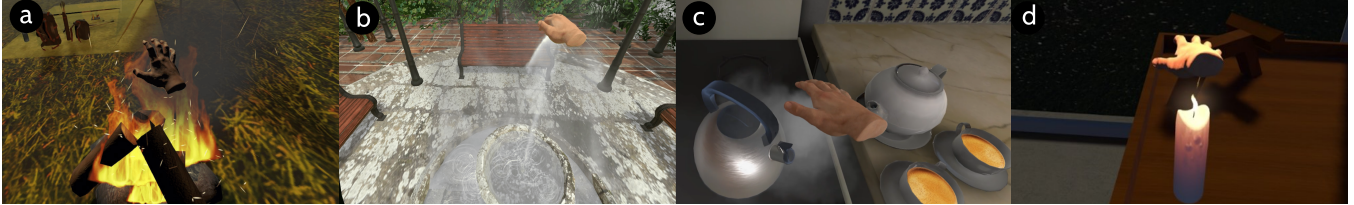


Figure 1: Users' point of view while interacting with (a) campfire scene, (b) water fountain scene, (c) kitchen scene, and (d) candle scene

ABSTRACT

We demonstrate an ultrasound haptic-based mid-air thermo-tactile display system, designed as a proof-of-concept with an open-top chamber, heat modules, and an ultrasound haptic display. Our method involves directing heated airflow toward the focused pressure point produced by the ultrasound display, delivering thermal and tactile cues in mid-air simultaneously. We showcase our system across four different VR environments—*campfire*, *water fountain*, *kitchen*, and *candle*—to illustrate the rich user experiences that result from integrating thermal and tactile feedback.

CCS CONCEPTS

• **Hardware** → **Emerging interfaces**; • **Human-centered computing** → **Haptic devices**.

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1 INTRODUCTION

In recent years, significant emphasis has been placed on incorporating thermal feedback in VR applications [Son et al. 2023; Wang et al. 2024] to enhance immersion and presence. Thermal feedback serves as an ambient communication channel, contributing to a more realistic perception of virtual objects and environments. Just as in the physical world, many VR scenarios benefit from thermal feedback—users can feel the warmth of a fire in a virtual fireplace,

experience the sensation of hot water from faucets or showers, or perceive the chill of falling snow.

Numerous studies on non-contact thermal and humidity feedback have employed various methods, utilizing thermal devices such as heaters, blowers, projector lights, and infrared lamps. For instance, Hülsmann et al. [Hülsmann et al. 2014] provided wind and thermal feedback using IR lamps and fans within the interaction space. Motoyama et al. [Motoyama et al. 2022] used mist to create cold sensations on the fingertip, while Kamigaki et al. [Kamigaki et al. 2020] proposed an ultrasound display to generate a focal point in mid-air on a hand-worn glove to deliver thermal sensations. However, these approaches often struggle to control thermal cues in 3D space due to their underlying mechanisms. Additionally, while these methods offer better control, they typically do not support bare-hand interaction.

Our perception-based approach integrates localized ultrasound tactile feedback with global thermal feedback, creating a thermo-tactile sensation on bare hands in mid-air. Tactile stimuli are perceived through mechanoreceptors and processed in the somatosensory cortex, while thermal stimuli are perceived through thermoreceptors and processed in the insula cortex. When these two types of stimuli are applied to the same area of the hands without masking each other, they are perceived as a unified thermo-tactile feedback. This allows for the presentation of thermo-tactile cues in 3D space at arbitrary locations. It combines vibrotactile and thermal stimuli without interference by generating ambient thermal cues through a heated chamber. By presenting acoustic pressure points alongside these ambient thermal cues, we achieve integrated thermo-tactile sensations in mid-air for bare-hand interaction. Visual information is closely coupled with these thermo-tactile cues (see Figure 1). Further details about this system and the experiments conducted to validate our approach can be found in [Singhal et al. 2021].

2 SYSTEM

2.1 Hardware

Figure 2 illustrates our setup that provides thermo-tactile feedback in mid-air. The system delivers tactile stimuli modulated at 200 Hz

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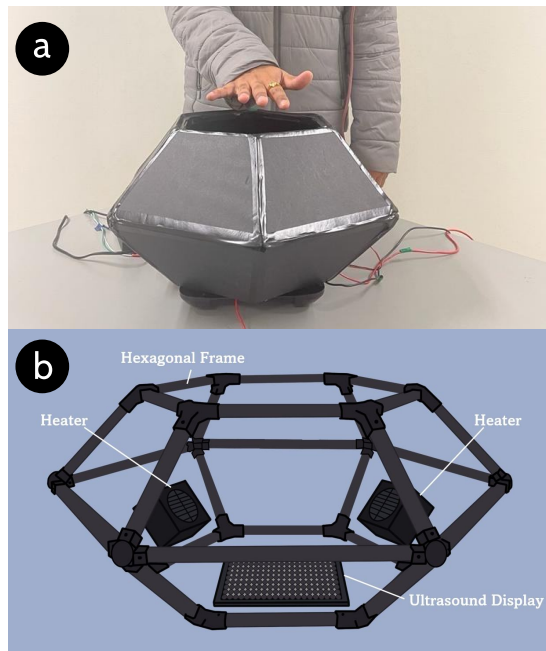


Figure 2: (a) A user interacting with the real system and (b) labeled diagram of the proposed system

to the user's palm using the ultrasound haptic display (STRATOS Explore, Ultraleap, UK), which can render various haptic patterns compatible with VR scenes. The user's hand is tracked using the Meta hand-tracking system. Thermal stimuli are delivered to the user's hand via a PTC (Positive Thermal Coefficient) heating element, which increases its resistance to maintain a constant temperature. The heater's temperature is controlled by a power supply. The chamber, designed in a hexagon shape to maintain thermal flow, is constructed with 30 aluminum beams for the frame and insulation foam for the surface cover. Temperature control is achieved by adjusting the digital power supply voltage to the PTC heater through a microcontroller. An Oculus Quest 2 VR headset is used in conjunction with our prototype system.

2.2 Visual Scenes

We demonstrate our approach using four VR scenes, each providing unique thermo-tactile feedback. We describe each environment in detail below:

CampFire: Figure 1 (a) shows the campfire scene. In this environment, the user interacts with fire embers and rising smoke with their virtual hands. An ultrasonic cue at 200 Hz with 0.7 intensity for haptic rendering is provided for 500 ms whenever an ember collides with the virtual hands. The temperature for this scene is set to 44°C, which maximizes the heat effect without reaching the pain threshold, simulating the warmth of a fire. A crackling fire sound is played to enhance the user experience.

Water Fountain: In this scene, a hot water fountain with a water stream is presented in a park (see Figure 1 (b)). When the water stream collides with the user's hands, the height of the water stream

dynamically adjusts depending on the hand's position. A continuous circular haptic sensation with a 2 cm radius, frequency of 200 Hz, and intensity of 1.0 is presented to the user's palm. A temperature of 38°C is selected to emulate the sensation of warm water hitting the palm.

Kitchen: In this scene, users find themselves in a kitchen with a teapot on the stove, emitting steam (see Figure 1 (c)). Participants can interact with the steam using their virtual hands. A haptic cue with a 4 cm radius, 200 Hz frequency, and 0.4 intensity is presented on the user's palm to simulate the sensation of steam interacting with the skin. The temperature for this scene is set to 40°C to replicate the warmth of steam.

Candle: In this scene, a candle with a bright flame is placed on a desk (see Figure 1 (d)). Participants can place their hands above the candle flame to perceive the fluctuating hot air. A circular haptic cue sensation with a radius changing from 2 cm to 0.5 cm over 2 seconds, a frequency of 200 Hz, and an intensity change from 1.0 to 0.8 is provided at 25 cm above the ultrasonic board. A temperature of 36°C is chosen to simulate the flickering of candle flame.

3 DEMO EXPERIENCE

At SIGGRAPH 2024 Emerging Technologies, we will showcase the proposed thermo-tactile display within a VR setup. Attendees interested in the interactive demo will receive a briefing on how to interact with the system. They will then stand in front of the system and put on the headset. Demonstrators will perform a calibration process to synchronize the attendee's physical and virtual world positions. Following calibration, attendees can begin interacting with the prototype, engaging with all the interactive objects. The entire demo, including calibration and briefing, will take no more than 5 minutes per attendee.

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